

Image Defect Detection and Localization Using Anomaly Detection

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Abstract

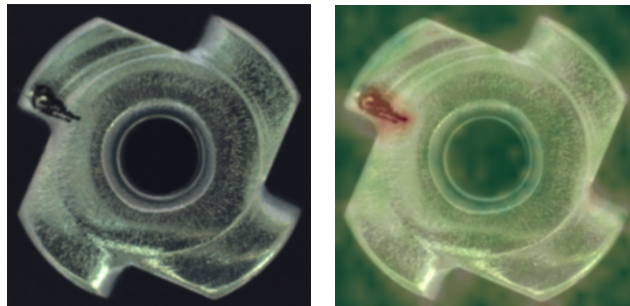
This work investigates the use of Machine Learning techniques for the automatic detection and localization of defects in images of industrial products. The proposed system uses a pre-trained ResNet-50 network for feature extraction, combined with the PatchCore method for anomaly detection. The approach is built using only defect-free images and is subsequently evaluated on both normal and anomalous samples. The objective is to correctly identify defective images and, through the generation of heatmaps, localize the regions most strongly associated with the presence of anomalies. The robustness of the system is further evaluated under domain shift, simulating realistic industrial acquisition variations such as illumination changes, sensor noise, and camera viewpoint shifts. To mitigate the resulting performance drop, the memory bank was enriched with augmented normal samples using softer versions of the same perturbations, and the effect of the amount of augmentation and of the classification threshold was also studied. The performance of the system is evaluated using standard binary classification metrics.

1. Introduction

The automatic detection of defects in images represents one of the most relevant computer vision application in industrial quality control, replacing slow and subjective manual inspection with fast, systematic anomaly detection.

In this work, the problem of *anomaly detection* applied to images of industrial products is addressed. The objective is to distinguish normal samples from defective ones and, when possible, localize the defect. Unlike a traditional supervised approach, the system is trained using only non-defective images, learning a representation of the normal appearance of the product and using this knowledge to identify deviations in test samples.

The proposed pipeline uses a pre-trained ResNet-50[3] as a feature extractor, combined with PatchCore[4], an anomaly detection method based on the comparison of local features. The representations extracted from intermediate layers of the network are used to build a *memory bank*



(a) Original image.

(b) Superimposed heatmap.

Figure 1: Example of defect detection and localization through anomaly detection.

containing the description of normal images only. During inference, distance metrics between test image features and the memory bank yield a global anomaly score and a spatial heatmap (shown in Fig. 1) to highlight defect locations. Model performance is evaluated using Accuracy, Precision, Recall, and F1-score alongside qualitative heatmap analysis.

System robustness is evaluated under *domain shift* conditions by introducing test-set alterations, including noise, blur, illumination changes, and geometric transformations. These alterations are then used to augment the memory bank to try and mitigate the the impact of domain shift on the performance of the system.

2. Dataset

For the development and evaluation of the proposed system, the MVTec Anomaly Detection (MVTec AD)[1] dataset was used. It is designed for industrial anomaly detection and contains high-resolution images of different object and material categories, including both defect-free and defective samples.

In this work, we focused on the *metal_nut* category, which contains images of metallic nuts. Focusing on a single category allows the model to learn a more specific representation of the normal appearance of the object.

The dataset follows the typical anomaly detection setting: the training set contains only *good* images, while the

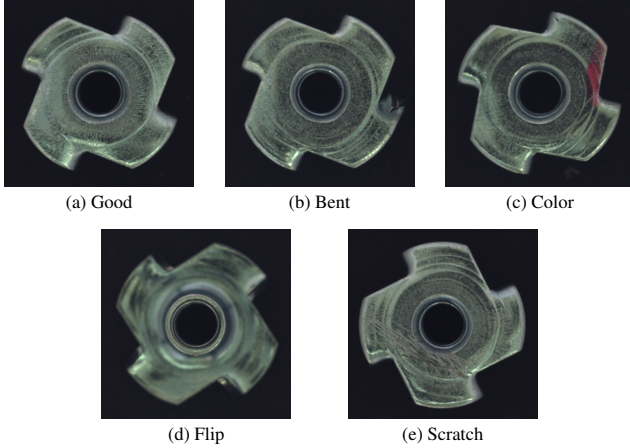


Figure 2: Examples from the *metal_nut* category of the MVTec AD dataset. Sample (a) represents a normal image, while (b)–(e) show different types of anomalies.

test set includes both normal and anomalous samples. The model therefore learns the normal appearance and detects deviations from it during testing.

Examples from the *metal_nut* category are shown in Fig. 2. The first image represents a normal sample, while the others show different types of defects.

2.1. Preprocessing

Before being processed by the ResNet-50 network all images undergo the same preprocessing pipeline, to obtain inputs with a consistent size and distribution, compatible with the pre-trained feature extractor.

The normalization uses the standard ImageNet statistics, $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$, for the RGB channels.

Table 1: Preprocessing pipeline applied to each image.

Operation	Parameters
Resize	256 pixels
Center Crop	224×224
Tensor Conversion	RGB image to tensor
Normalization	ImageNet statistics

The same preprocessing pipeline is applied to both training and test images. This ensures that the features stored in the memory bank and those extracted during inference are generated under the same input conditions.

2.2. Dataset Organization

The images are organized into training, validation, and test sets. The original MVTec AD training set for the *metal_nut* category contains 220 defect-free images. These samples are randomly divided into two subsets: 80% of

the images are used to construct the normal memory bank, while the remaining 20% are used as a validation set for threshold selection.

Using a fixed random seed, this results in 176 training images and 44 validation images. Both subsets contain exclusively defect-free samples.

The test set contains both *good* samples and images affected by different anomaly types. During evaluation, all defective categories are considered anomalous, resulting in a binary classification problem between normal and anomalous samples.

The validation images are never included in the memory bank, ensuring that the anomaly threshold is estimated using normal samples that were not used to construct the reference feature representation.

Table 2: Number of images used for training, validation, and testing.

Subset / Category	Images
Good (Training)	176
Good (Validation)	44
Good (Test)	22
Bent	25
Color	22
Flip	23
Scratch	23
Total	335

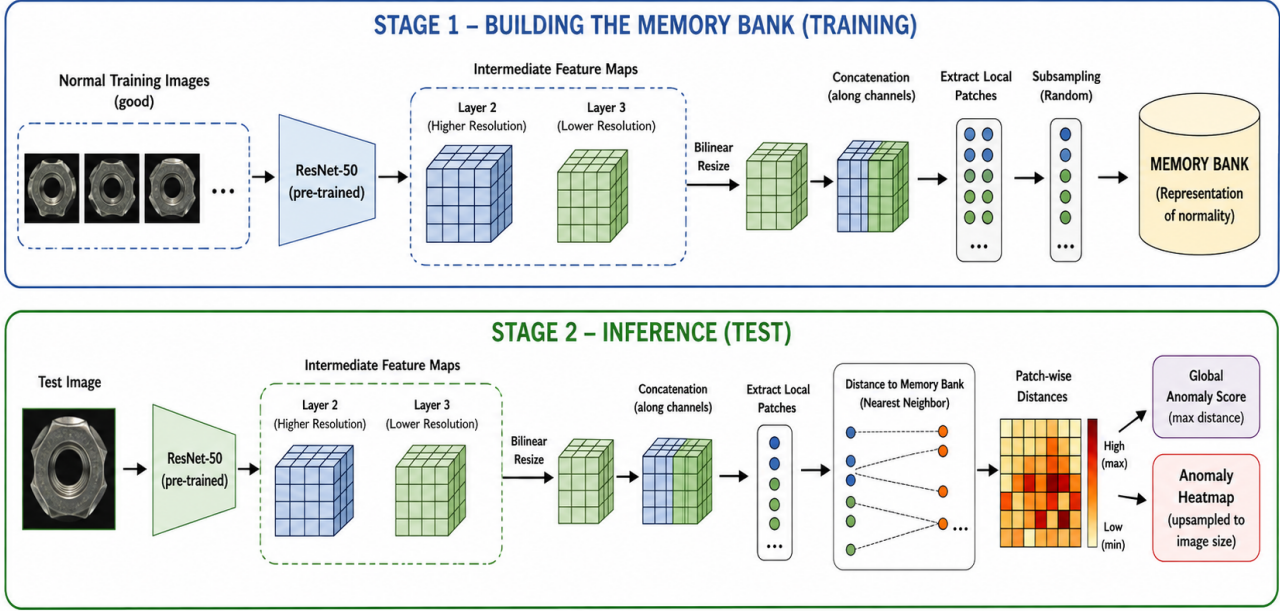
Table 2 reports the number of images available for each category. The normal samples are separated between training and test sets, while the anomalous test samples are divided according to their specific defect type.

3. Methodology

The proposed anomaly detection system combines a pre-trained ResNet-50 with a PatchCore-based feature comparison approach. The pipeline consists of two stages: building a representation of normal samples and evaluating new test images.

During training, only defect-free images are processed by ResNet-50. Intermediate feature maps are converted into local feature vectors and stored in a *memory bank*, representing the normal appearance of the object.

During inference, the same features are extracted from each test image. Each local vector is compared with the memory bank, and the distance to the nearest normal feature is used as the anomaly measure. These distances are used to compute a global anomaly score and generate a spatial anomaly map for approximate defect localization.



Overall pipeline of the proposed method. Stage 1 builds the memory bank using only normal training images. Stage 2 computes patch-wise distances for a test image to obtain the global anomaly score and the anomaly heatmap.

3.1. Feature Extraction with ResNet-50

A ResNet-50 model pre-trained on ImageNet[2] is used as the feature extractor. Since the objective is not to perform conventional image classification, the final classification output of the network is discarded. Instead, intermediate representations from *layer2* and *layer3* are collected through forward hooks.

The parameters of ResNet-50 are frozen during the entire procedure, preventing their modification and allowing the network to be used exclusively as a feature extractor. This also avoids the need to train a deep neural network from scratch using the relatively limited number of normal samples available for the selected object category.

The two selected layers provide feature maps at different levels of abstraction and spatial resolution. Features extracted from *layer2* retain more spatial information, while those obtained from *layer3* provide a higher-level representation of the image content.

Since the feature maps produced by the two layers have different spatial dimensions, the output of *layer3* is resized through bilinear interpolation to match the spatial resolution of *layer2*. The resulting tensors are then concatenated along the channel dimension, combining information from both levels of the network.

3.2. Memory Bank Construction

For each normal training image, the combined feature tensor has dimensions

$$B \times C \times H \times W, \quad (1)$$

where B represents the batch size, C the number of feature channels, and H and W the spatial dimensions of the feature map.

The tensor is rearranged so that each spatial location is represented by an independent feature vector of dimension C . Therefore, each image contributes $H \times W$ local descriptors to the memory bank.

The complete memory bank can be represented as

$$\mathcal{M} = \{m_1, m_2, \dots, m_N\}, \quad (2)$$

where each $m_i \in \mathbb{R}^C$ represents a local feature extracted from a defect-free training image.

The memory bank is constructed exclusively from the 176 normal images belonging to the training subset. The 44 validation images are not included in this process and are used only for threshold estimation.

Since the complete set of extracted patches is large, random subsampling is applied to reduce memory requirements and computational cost. In the final configuration, 20% of the available patch descriptors are retained in the memory bank.

3.3. Anomaly Score

During inference, a test image is processed using the same ResNet-50 feature extraction pipeline. After resizing and concatenating the feature maps from *layer2* and *layer3*,

the resulting representation is converted into a set of test patches

$$\mathcal{P} = \{p_1, p_2, \dots, p_K\}. \quad (3)$$

For each test patch p_i , the Euclidean distance from all elements of the memory bank is computed. The distance from the closest normal feature is then selected:

$$d_i = \min_{m_j \in \mathcal{M}} \|p_i - m_j\|_2. \quad (4)$$

A low value of d_i indicates that the patch is similar to a region observed in the normal training images, while a high value indicates a stronger deviation from the learned normal representation.

The global anomaly score assigned to the entire image is obtained by selecting the maximum patch distance:

$$S(x) = \max_i d_i. \quad (5)$$

The final classification threshold is estimated using the normal validation set. After constructing the memory bank, an anomaly score is computed for each validation image. The threshold T is then defined as the 90th percentile of the resulting score distribution:

$$T = P_{90}(\{S(x_v)\}_{x_v \in \mathcal{V}}), \quad (6)$$

where $\mathcal{V}$ denotes the validation set. A test image is classified as anomalous when $S(x) > T$.

3.4. Anomaly Localization

The same patch distances used for image-level classification are also exploited to provide an approximate localization of the defect. Since every distance d_i corresponds to a spatial location in the intermediate feature map, the values can be rearranged into a two-dimensional anomaly map.

The resulting map is normalized and resized to match the spatial resolution of the input image. A color map is then applied and superimposed on the original image, producing a heatmap in which regions associated with larger feature distances are visually highlighted.

This representation provides additional information compared with the global anomaly score, since it allows the system to indicate which areas of the image contributed most strongly to the anomaly prediction.

4. Implementation

The proposed system was implemented in Python using PyTorch as the main deep learning framework. The implementation was divided into separate modules responsible for dataset management, feature extraction, anomaly detection, evaluation, and visualization. This modular organization simplifies the execution of the different stages of the

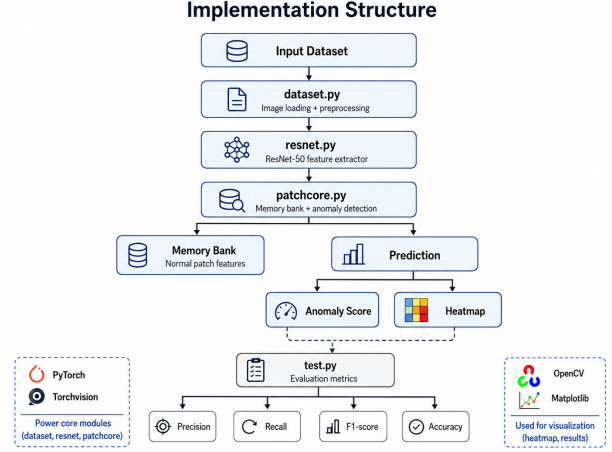


Figure 3: Software organization of the implemented anomaly detection system. The dataset module handles image loading and preprocessing, the ResNet-50 module extracts intermediate features, the PatchCore-based module constructs the memory bank and performs anomaly detection, while the evaluation module computes the final performance metrics.

pipeline and allows each component to be modified independently.

The main external libraries used in the implementation include *torch* and *torchvision* for neural network operations, *PIL* for image loading, *NumPy* for numerical operations, *Scikit-learn* for auxiliary feature processing, *OpenCV* for image resizing, and *Matplotlib* for the visualization and generation of anomaly heatmaps.

4.1. Model and Feature Extraction

The feature extractor is implemented as a custom PyTorch module containing a ResNet-50 model initialized with ImageNet pre-trained weights. All network parameters are frozen by setting their gradient computation to disabled, since no fine-tuning is performed.

Forward hooks are registered on *layer2* and *layer3* in order to store their outputs during a forward pass. The final classification output of ResNet-50 is therefore ignored, while the intermediate feature maps are returned and subsequently processed by the anomaly detection module.

The PatchCore-based component receives the feature extractor and the execution device during initialization. During memory bank construction, batches of normal images are processed without gradient computation. The extracted feature maps are transferred to the CPU, resized when necessary, concatenated, and reshaped into local feature vectors.

The implementation also provides a configurable sub-sampling operation to reduce the number of vectors stored

in the memory bank. A random subset of the available patch representations is retained according to a predefined percentage.

4.2. Inference and Classification

During inference, each test image is processed using the same preprocessing pipeline adopted for the training samples. An additional batch dimension is introduced before the image is provided to the feature extractor.

The extracted test patches are compared with the memory bank using the `torch.cdist` function, which computes pairwise Euclidean distances. For each test patch, only the minimum distance from the normal reference features is retained. The maximum value among these minimum distances represents the global anomaly score of the image.

The classification threshold is not manually fixed. After the memory bank has been constructed, the 44 normal validation images are independently processed by the model and their anomaly scores are collected. The final threshold is selected as the 90th percentile of this distribution.

This procedure provides a threshold derived exclusively from normal samples that were not used for memory bank construction. Once determined, the threshold is kept fixed during the evaluation of both the original test set and the domain shift experiments.

An image is classified as anomalous when its anomaly score is greater than T ; otherwise, it is classified as normal.

The main implementation parameters are summarized in Table 3.

Table 3: Main parameters of the implemented anomaly detection system.

Parameter	Value
Backbone	ResNet-50
Pre-trained weights	ImageNet
Input resolution	224×224
Extracted layers	Layer 2, Layer 3
Feature interpolation	Bilinear
Distance metric	Euclidean
Train / validation split	80% / 20%
Memory bank reduction	Random subsampling (20%)
Threshold selection	Validation 90th percentile

4.3. Evaluation

The evaluation procedure iterates over all categories contained in the test set. Samples belonging to the *good* category are assigned the normal label, while all defect categories are considered anomalous.

For every test image, the predicted class is compared with the corresponding ground-truth label. The resulting

predictions are used to compute the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Based on these values, the performance of the system is evaluated using Accuracy, Precision, Recall, and F1-score.

Accuracy measures the overall proportion of correctly classified samples, *precision* measures the proportion of samples classified as anomalous that are actually defective, *recall* measures the proportion of defective samples that are correctly identified by the system while the *F1-score* represents the harmonic mean between Precision and Recall.

These metrics together provide a complete picture of classification performance, capturing both overall correctness and the trade-off between false positives and false negatives.

The same threshold obtained from the normal validation set is used for all test samples. No information from the test labels is used during threshold selection.

4.4. Heatmap Generation

For anomalous test images, the system can additionally generate a visual representation of the localization result. The patch-level anomaly distances are first reshaped into a two-dimensional map and normalized between 0 and 1.

The anomaly map is then resized to the same spatial dimensions as the processed input image using OpenCV. A color map is applied to the normalized values and the resulting heatmap is superimposed on the original image.

In the implemented visualization, the final output is obtained by combining 60% of the original image with 40% of the colored anomaly map. The resulting figures contain both the original sample and its corresponding anomaly visualization, together with the global anomaly score computed by the model.

5. First part Results

The proposed anomaly detection system was evaluated on the test portion of the *metal_nut* category. Before testing, the 44 normal validation images were processed to estimate the classification threshold. The threshold was selected as the 90th percentile of the validation anomaly score distribution. The resulting threshold was then kept fixed for the complete test set. The quantitative evaluation is based on Accuracy, Precision, Recall, and F1-score, while the generated heatmaps are used for qualitative localization analysis.

The resulting threshold was $T = 36.02$.

5.1. Overall Classification Performance

The overall classification results obtained on the complete test set are reported in Table 4.

The high Precision indicates that most samples classified as anomalous were actually defective. At the same time, the

Table 4: Overall classification performance on the *metal_nut* test set.

Metric	Value
Accuracy	91.3%
Precision	96.6%
Recall	92.5%
F1-score	94.5%

Table 5: Confusion matrix of the anomaly detection system.

	Predicted Normal	Predicted Anomaly
Actual Normal	TN = 19	FP = 3
Actual Anomaly	FN = 7	TP = 86

Recall above 92% shows that the system successfully detected the large majority of defective samples. The resulting F1-score confirms a good balance between false alarms and missed defects.

5.2. Confusion Matrix

A more detailed representation of the classification results can be obtained through the confusion matrix. Table 5 reports the number of True Positives, False Positives, True Negatives, and False Negatives produced by the system.

The confusion matrix shows that 19 of the 22 normal test images were correctly classified, while 3 normal samples were incorrectly identified as anomalous. Among the 93 defective images, 86 were correctly detected and 7 were incorrectly classified as normal.

The relatively small number of False Positives is consistent with the high Precision obtained by the system. False Negatives are particularly relevant in an industrial quality-control scenario, since they correspond to defective products that would incorrectly be considered acceptable. Nevertheless, the model successfully detected more than 92% of the anomalous samples.

5.3. Qualitative Localization Results

In addition to the numerical evaluation, the anomaly localization capability of the system was analyzed through the generated heatmaps. Some representative examples are shown in Fig. 4.

The qualitative results make it possible to verify whether the regions associated with high anomaly scores correspond to the actual defective areas. A successful localization is obtained when the highest values of the heatmap are concentrated around the visible defect, while low anomaly values are assigned to the remaining parts of the object.

In the selected examples, the regions associated with the highest anomaly values are concentrated around the visible defects, while most of the defect-free surface presents lower

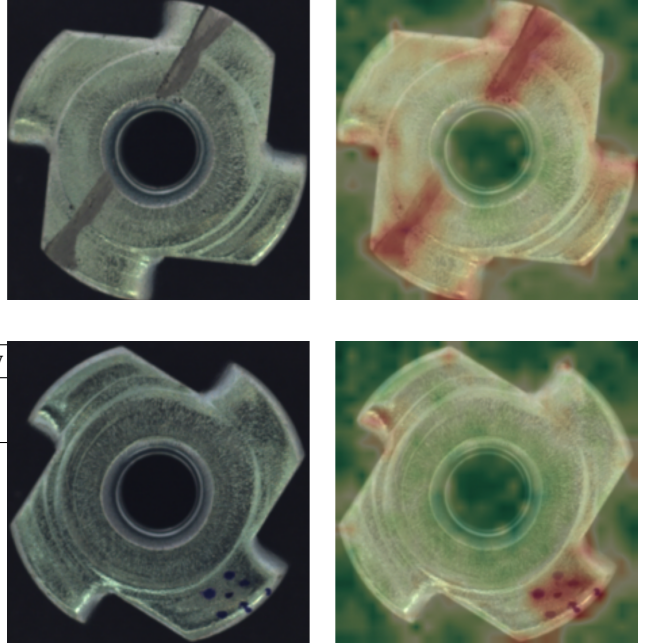


Figure 4: Examples of qualitative anomaly localization results. For each sample, the original image is compared with the corresponding heatmap generated from the patch-level anomaly distances. Regions associated with higher anomaly values are highlighted more strongly.

Table 6: Summary of the anomaly scores obtained on the test set.

Class	Mean Score	Std. Dev.
Normal	33.70	2.58
Anomalous	43.00	4.14

anomaly responses. This indicates that the patch-level distances used for classification also preserve useful spatial information for approximate defect localization.

5.4. Anomaly Score Analysis

The anomaly scores obtained on the test set show a clear difference between normal and defective samples. Normal images obtained an average score of 33.70, while anomalous images reached an average value of 43.00.

The difference between the two mean scores indicates that the feature representation stored in the memory bank provides a useful separation between normal and anomalous samples. The selected threshold of $T = 36.02$ lies between the average scores of the two classes and allows most defective images to be detected while limiting the number of False Positives.

However, the non-zero classification errors indicate that the two score distributions are not completely separated.

Table 7: Perturbation models used for domain shift evaluation.

Transform	Model	Range
Exposure	$I' = \alpha I + \beta$	$\alpha \in [0.5, 1.7]$
White balance	per-channel scaling (R, B)	$[0.70, 1.40]$
Noise	Gaussian + salt-and-pepper	$\sigma \in [5, 40]$
Contrast	CLAHE on L channel (LAB)	clip $\in [2, 6]$
Perspective	homography from tilt	$\pm 15^\circ$
Specular	additive Gaussian hotspot	radius $\in [0.05, 0.25]$
Gamma	$I' = 255 \left(\frac{I}{255}\right)^\gamma$	$\gamma \in [0.45, 2.2]$
Affine	translation + rotation + shear	$t_x, t_y \in [-40, 40]$ px $\theta \in [-25, 25]^\circ$ $s \in [-0.08, 0.08]$

Some defective samples produce scores similar to normal images, while a small number of normal samples exceed the selected threshold.

6. Domain Shift

In real industrial environments, acquisition conditions may differ from those used to construct the normal memory bank. Variations in illumination, sensor response, image quality, or camera viewpoint can modify the extracted features even when no actual defect is present. To evaluate the robustness of the proposed system, several perturbations were therefore applied to the original test images while preserving their ground-truth labels.[5]

6.1. Perturbations and Evaluation Protocol

Six transformations were considered for the robustness evaluation: *exposure*, *white balance*, *noise*, *contrast*, *perspective*, and *specular*. The other two were not used in the next mitigation step except for testing the results of it.

The transformations simulate realistic acquisition variations: exposure models illumination changes, white balance models different lighting conditions, noise represents sensor noise, contrast models camera tone-curve differences, and perspective simulates a slight camera tilt. Specular highlights model glare on the metallic surface.

Representative examples are shown in Fig. 5.

For each test image, the parameters of the selected transformation are randomly sampled independently within the corresponding ranges in Table 7. The shifted images are used only during evaluation: the feature extractor, memory bank, and anomaly threshold remain unchanged. Accuracy, Precision, Recall, and F1-score are computed for each condition.

6.2. Anomaly Localization under Domain Shift

In addition to classification performance, the anomaly maps were inspected to assess whether domain shifts also affect spatial anomaly localization. Despite the applied perturbations, the generated heatmaps generally continued to highlight the defective regions of the test images. This sug-

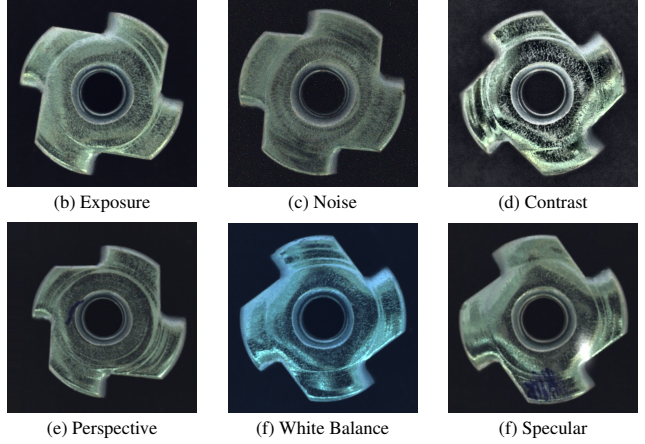


Figure 5: Representative domain shift transformations applied to the test images.

gests that the local feature distances retain useful spatial information even when the input appearance differs from the normal samples stored in the memory bank.

Figure 6 shows representative examples under different perturbations. Although some shifted images produce additional responses in non-defective regions, the main defective areas remain clearly highlighted, indicating that the proposed approach provides reasonably robust approximate localization under domain shift.

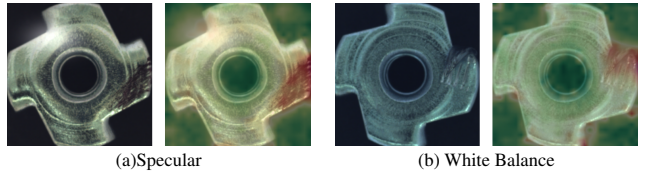


Figure 6: Representative anomaly localization results under domain shift. The heatmaps continue to highlight the defective regions despite changes in image appearance.

6.3. Robustness Mitigation

To address the performance drop caused by the perturbations, the memory bank was enriched with augmented normal samples: for each of the 176 training images, one of the five perturbations was applied at random, and the resulting image was added to the memory bank alongside the original. Using more than one augmented copy per image (2 or 3) was also tested, but consistently gave lower performance than a single copy. T was varied across $\{85, 87.5, 90, 92.5, 95\}$. Lowering it to the 85th percentile slightly increased Recall on some conditions, but reduced Accuracy and Precision a bit too much across the board, including on the original, unshifted test set. The 90th percentile, used throughout this work, remained the best over-

all choice and was kept unchanged.

6.4. Results and Discussion

The baseline robustness results, without memory bank augmentation, are summarized in Table 8. Contrast and Perspective show the largest drops in Accuracy and Precision compared with the nominal condition.

Table 8: Classification performance under domain shift.

Condition	Acc.	Prec.	Rec.	F1
Original	91.3%	96.6%	92.5%	94.5%
Exposure	89.0%	91.6%	95.0%	93.3%
White Balance	89.0%	89.6%	97.5%	93.4%
Noise	90.0%	91.6%	96.3%	93.9%
Contrast	85.0%	84.2%	100%	91.4%
Perspective	85.0%	87.4%	95.0%	91.0%
Specular	89.0%	90.6%	96.3%	93.3%
Gamma	87.0%	87.6%	97.5%	92.3%
Affine	83.0%	82.5%	100%	90.4%
Average (shifted)	87.8%	89.2%	96.7%	92.7%

After memory bank augmentation, performance improves across most shifted conditions, as reported in Table 10. Average Accuracy increases from 87.8% to 91.7%, Precision from 89.2% to 96.0%, and F1-score from 92.7% to 94.7%. Recall decreases slightly from 96.7% to 93.6%, indicating a trade-off between anomaly detection and false-positive reduction.

Table 9: Classification performance under domain shift after memory bank augmentation.

Condition	Acc.	Prec.	Rec.	F1
Original	93.0%	98.9%	92.5%	95.6%
Exposure	89.0%	92.6%	93.8%	93.2%
White Balance	93.0%	97.4%	93.8%	95.5%
Noise	92.0%	97.4%	92.5%	94.9%
Contrast	94.0%	95.1%	97.5%	96.3%
Perspective	91.0%	97.3%	91.3%	94.2%
Specular	91.0%	96.1%	92.5%	94.3%
Average (shifted)	91.7%	96.0%	93.6%	94.7%

Table 10: Classification performance under domain shift after mild (0.6) memory bank augmentation.

Condition	Acc.	Prec.	Rec.	F1
Original	92.2%	97.7%	92.5%	95.0%
Exposure	90.0%	92.6%	93.8%	93.2%
White Balance	93.0%	96.1%	91.3%	93.6%
Noise	88.0%	97.2%	87.5%	92.1%
Contrast	92.0%	90.9%	100%	95.2%
Perspective	90.0%	96.1%	91.3%	93.6%
Specular	94.0%	97.4%	95.0%	96.2%
Gamma Correction	86.0%	92.3%	90.0%	91.1%
Affine	87.0%	86.8%	98.8%	92.4%
Exposure + Perspective	87.72%	87.5%	94.6%	90.9%
Average (shifted)	91.2%	95.1%	93.2%	94.0%

To evaluate generalization beyond the transforms used for augmentation, the mitigated memory bank was also tested on Gamma Correction, Affine transformation, and a

combined Exposure + Perspective condition, none of which were included during augmentation. The results indicates that the augmented memory bank generalizes to some unseen shift types more than others, rather than uniformly.

Increasing the number of augmented samples was also tested, but performance consistently decreased. This suggests that excessive augmentation introduces unnecessary variability into the normal feature distribution; therefore, a single augmented sample per training image was retained.

7. Conclusion

The system was evaluated on the *metal_nut* category of the MVTec AD dataset, and under nominal conditions, the baseline system achieved 91.3% Accuracy, 96.6% Precision, 92.5% Recall, and 94.5% F1-score. The generated heatmaps also showed that the main defective regions could be approximately localized from the patch-level anomaly scores.

Robustness was then evaluated under domain shift, the average performance across shifted conditions was 87.8% Accuracy, 89.2% Precision, 96.7% Recall, and 92.7% F1-score. To mitigate this degradation, augmented versions of the normal training images were added to the memory bank. This increased the average Accuracy to 91.7%, Precision to 96.0%, and F1-score to 94.7%, while Recall decreased slightly to 93.6%. The largest improvements were observed for Contrast and Perspective. The anomaly maps also remained qualitatively useful under the shifted conditions, although some additional responses appeared in non-defective regions.

A further experiment focused on controlling the strength of the memory bank augmentation. The parameter AUGMENTATION_SCALE=0.6 was introduced to reduce each perturbation range to 60% of its original width around its midpoint, producing softer versions of the test-time transformations.

Finally, generalization beyond the perturbations used for augmentation was evaluated, with mixed results. These experiments suggest that memory bank augmentation can also provide a degree of generalization to previously unseen acquisition variations, but it depends on the transformation.

Overall, the results demonstrate that enriching the memory bank with perturbed normal samples enhances robustness to domain shift while preserving effective anomaly localization. Because a trade-off inherently exists between recall and other metrics, we do not propose a single optimal configuration. Instead, the final deployment parameters must be carefully calibrated to align with the specific operational requirements and priorities of the target industrial environment.

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